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REGULAR ARTICLE

Using difference equations to find optimal tax structures on the SugarScape

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Abstract The use of equations to describe agent-based model dynamics allows access to mathematical theory that is not otherwise available. In particular, equation models can be effective at solving optimization problems—that is, problems concerning how an agent-based model can be most effectively steered into a particular state. In order to illustrate this strategy, we describe a modified version of the well-known SugarScape model and implement taxation. The optimization problem is to determine tax structures that minimize deaths but maximize tax income. Tax rates are dependent upon the amount of sugar available in a particular region; the rates change over time. A system of discrete difference equations is built to capture agent-based model dynamics. The equations are shown to capture the dynamics very well both with and without taxation. A multi-objective optimization technique known as Pareto optimization is then used to solve the problem. Rather than focusing on a cost function in which the two objectives are assigned weights, Pareto optimization is a heuristic method that determines a suite of solutions, each of which is optimal depending on the priorities of the researcher. In this case, Pareto optimization allows analysis of the tradeoff between taxes collected and deaths caused by taxation. The strategies contained here serve as a framework for a broad class of models.

Keywords Optimization · SugarScape · Approximation · Difference equations

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1 Introduction

In many agent-based models (ABMs), even the simplest local interactions can give rise to complicated and unpredicted global dynamics, indicating both the power of agent-based modeling and the care that must be taken in any attempted analysis. A mathematical description of an ABM has particular appeal because it brings the desired rigor to the form of the model, and allows access to a plethora of theory and tools. For many ABMs, questions amenable to investigation arise naturally: for example, what is the best investment strategy in order to maximize profit? How should resources be allocated in order to maximize customer satisfaction? What is the ideal tax policy for a given scenario? We refer to these as *optimization* problems, because they are focused on determining the optimal way to achieve a particular goal.

There are methods for solving optimization problems for ABMs directly—this area of study is referred to as simulation optimization. However, the approximation of ABM dynamics by mathematical equations is another method for solving optimization problems, and one which brings to light another advantage: rather than relying on results from simulation—which can be computationally expensive and unreliable—the equation systems are rigorous and can be evaluated very quickly. Additionally, if the equations are analytical rather than phenomenological, the solving of optimization problems can offer additional insight into model dynamics, as results can be interpreted in terms of the model parameters.

To illustrate this technique, we introduce a simplified version of a well-known ABM, pose an optimization problem focused on the determination of optimal tax structures, and develop analytical equations to capture pertinent model dynamics. Finally, we use the equations to solve the optimization problem and then verify that the solutions faithfully translate back to the ABM. The results indicate the power of this approach, as well as informing directions for future research.

1.1 Related work

Many studies have examined similarities and differences between ABMs and equation-based models (EBMs). In their user's guide, Parunak et al. (1998) contrast ordinary differential equation (ODE) models with ABMs, concluding that ABMs are more appropriate in models with highly local features. Fahse et al. (1998) use an ABM of birds in an attempt to extract a population growth rate from ABM data; a similar goal is examined in a model by Duboz et al. (2003). A spatially heterogeneous ABM is used to derive a series of EBMs in a study by Picard and Franc (2001); here, local spatial effects are shown to be critically important. Mean-field aggregations of ABM dynamics are shown to work well by Edwards et al. (2003), while Ovaskainen and Cornell (2006) emphasize the importance of local densities in such models. Huet et al. (2007) examine the issue of local neighborhoods in the development of mean-field approximations. A broader study on the use of ABMs versus EBMs discusses circumstances for which a particular model type is preferable (Rahmandad and Sterman 2008). Cecconi et al. (2010) use a model from game theory to generate empirical data in order to fit an EBM

to that data; they emphasize the complementary nature of the two modeling paradigms. A formal methodology for translation of ABMs to polynomial dynamical systems (PDS) over a finite field is presented by Hinkelmann et al. (2011); this technique demonstrates that in theory, every ABM can be represented as a PDS.

In addition to general studies on agent- and equation-based modeling, simulation optimization techniques have been investigated in order to solve problems similar to those posited in Sect. 1. Okell et al. (2008) examine the effects of therapy and treatment in an ABM of malaria transmission, while Kasaie et al. (2010) focus on optimal resource allocation. Strategies for mitigating influenza outbreaks are the focus of studies by Mao (2011) and Yang et al. (2011). A textbook chapter by Laubenbacher et al. (2013) examines optimal control theory for ABMs, offering examples of simulation optimization strategies.

In this study, a system of discrete difference equations is developed analytically from ABM rules; the equations are subsequently used to solve an optimization problem. While optimal control theory exists for the mathematical system, it is often limited to cases of very few equations or a very short time horizon. Ding et al. (2007) describe such theory for a model containing several equations, but simulations are limited to fewer than ten time steps. Lenhart and Workman (2007) dedicate a textbook chapter to the topic but also focus on small systems. An infinite-time discrete optimal control problem is investigated by Hayek (2011); it is not clear if results are generalizable to finite-time systems.

Existing optimal control theory is intractable for models containing many equations over a long (but finite) time horizon, and translation of spatially heterogeneous ABMs often results in precisely these types of systems. Hence, heuristic methods are employed in order to solve optimization problems. Heuristic methods have been used in conjunction with ABMs to solve optimization problems related to cancer vaccination (Lollini et al. 2006; Pennisi et al. 2008) and HIV treatment (Castiglione et al. 2007), for example. Different heuristic methods for determining general strategies are contrasted by Pappalardo et al. (2010). Hence, there is precedent for the use of such methods in solving the same optimization problem in an equation-based model.

2 The model

In order to illustrate how analytical difference equations can be used to describe agent-based model dynamics, a representative model is used as a running example. The version of the model used here is based on SugarScape (Epstein and Axtell 1996), a large-scale ABM wherein virtual ants patrol a landscape made up of sugar. SugarScape was chosen as a representative model because it was the first large-scale ABM and it can be studied in a variety of contexts. Although first introduced as an abstract study of generative social science, SugarScape has been used to study economic trade (Dascàlu et al. 1998) and wealth distribution (Rahman et al. 2007); it even served as the basis for a study on the culture and migratory patterns of the Anasazi people (Dean et al. 2000). We use a modified version of the model: while simplified in order to highlight how analytical equations can be built, our version maintains spatial heterogeneity and complicated local dynamics. In the following section we provide a brief description

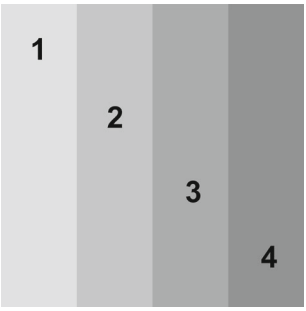

Fig. 1 Landscape of the SugarScape gradient model; *labels* indicate the amount of sugar contained on each grid cell in that region

of the model; full details (including parameter values and pseudo-code) are provided in "Appendix 1". It should be noted that there are many free software platforms for agent-based modeling, including Repast (North et al. 2006), MASON (Luke et al. 2005), and NetLogo (Wilensky 2009). A comparison of these platforms is outside the scope of this text. Results for this study were generated using NetLogo—in fact, the SugarScape Gradient model is adapted from a model contained in the NetLogo model library—though by using the description in "Appendix 1", the model can be replicated in a variety of agent-based modeling platforms. NetLogo was chosen for its intuitive interface and ease of use.

2.1 The SugarScape gradient model

The landscape in the version of SugarScape studied here consists of four vertical regions, increasing in sugar from left to right (see Fig. 1); hence, we refer to this as the SugarScape Gradient model. While edges wrap vertically, they are bound horizontally, so ants can only move to a higher sugar region by moving to the right. Each ant has its own metabolism, determining how much sugar it burns each time step. Ants collect sugar at their current location and lose sugar based on their metabolism; an ant dies if it runs out of sugar. In addition to the modified landscape, there are several changes to the original SugarScape model rules that bear mentioning:

- Ants can collect up to a **maximum** of 50 sugar,
- upon being collected by an ant, the sugar on that grid cell is **not** depleted,
- multiple ants may occupy one grid cell at any given time,
- ants move to the grid cell within their vision with the most sugar; if there are multiple such locations, ants choose one **at random**.

In order to make the analytical structure of the equations clearer, all ants have metabolism of 1 or 2, and vision 1 (meaning they can see one grid cell in each of the four principal directions).

2.2 An optimization problem

In the SugarScape Gradient model, wealth (i.e., sugar) is of natural interest: it drives the migration of the ants and determines their survival. In order to pose an optimization

problem, we impose *taxation* in the model: at regular intervals, ants are taxed a proportion of their sugar, depending on their location at the time of taxation. The purpose of taxation is to study its effect on the overall population dynamics: if taxes are always higher in a particular region, will agents tend to avoid that location? How high can taxes be before all of the ants die as a result? There are many optimization questions that arise from this line of thinking. We pose the following: what is the optimal tax structure in order to minimize the number of deaths and maximize the amount of tax collected? There are four regions, and five permissible tax rates: $\{0,\ 0.25,\ 0.5,\ 0.75,\ 1\}$. Tax is collected ten times, and the rates for each region may change each time it is collected. Hence there are $5^{4\cdot 10} \approx 9 \times 10^{27}$ possible tax structures. This is far too many for enumeration via simulation, highlighting the importance of both the equation system and the heuristic approach.

3 Analytical equations

In order to derive analytical equations describing ABM dynamics, it is important to consider the optimization problem. Since our optimization problem involves taxation, we need information about the location of the ants (since tax is location dependent) and the wealth distribution (so we know how much income we collect). Furthermore, the equation system is being used to find optimal tax structures for the ABM; hence it is preferable that a given tax structure can be implemented without alteration in both the ABM and the equation system. Hence, results obtained from the equations can be tested directly in the ABM without any modification. Several example equations are explained here; for the full set of equations see "Appendix 2". It should be noted that in the following explanations, 'wealth' and 'sugar level' are used interchangeably.

3.1 Population equations

Each of the four regions consists of 12 columns; since the map wraps in the vertical direction each column is homogeneous. Hence there are $4 \cdot 12 = 48$ population equations $p_1, \ldots, p_{48}$ tracking the number of ants in each column at each time step. We use discrete-time difference equations whose values are updated according to values at the previous time step. Consider the equation for $p_5(t)$, given in Eq. 1. The left half of Fig. 2 illustrates this equation: each ant can see exactly 5 grid cells, and in this case all of them have the same amount of sugar (since the ants are in the interior of the region). We expect 20 % of the ants to move to the left, 60 % to stay in their current column, and 20 % to move to the right (note that we do not need to track the vertical location of the ants). Note that 20 % of the ants in column 5 move to column 4 and the remaining 20 % move to column 6, accounting for the entire population in column 5. This can be seen more clearly by examining the full set of equations in "Appendix 2".

$$p_5(t+1) = 0.2p_4(t) + 0.6p_5(t) + 0.2p_6(t). \tag{1}$$

If an ant is at the rightmost edge of a boundary, it can see into the next region and will consequently move there. At the leftmost edge ants can see a column with lower

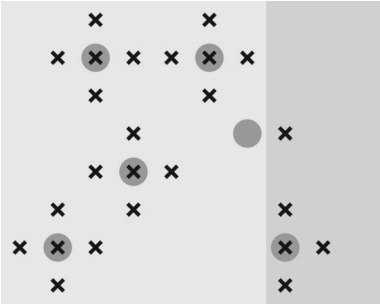

Fig. 2 Movement is determined by sugar levels within an ants vision

sugar than their current location; in that case, they are guaranteed to **not** move there. The right half of Fig. 2 illustrates these scenarios; Eqs. 2 and 3 show examples of how boundary column equations account for these effects.

$$p_{12}(t+1) = 0.20p_{11}(t), \tag{2}$$

$$p_{13}(t+1) = p_{12}(t) + 0.75p_{13}(t) + 0.2p_{14}(t). \tag{3}$$

Similar equations hold for each of the remaining columns; for the full set of equations see "Appendix 2".

3.2 Wealth distribution

Ants may accumulate up to 50 sugar; ants with 0 sugar die. Some ants have metabolism 1 and some have metabolism 2, and these are tracked separately. Finally, since taxation occurs by region, wealth distributions need to be tracked by region as well. Hence, since there are 51 sugar levels (0–50) for ants at each metabolism, there are $51 \cdot 2 = 102$ wealth equations per region, for a total of 408 wealth equations. Let $w_{r,m,s}(t)$ represent the number of ants in region r with metabolism m and sugar level s. The net sugar gain is $r - m$, so in general,

$$w_{r,m,s}(t+1) = w_{r,m,s-(r-m)}(t). \tag{4}$$

Some modification is needed for particular sugar levels in each region: for example, ants cannot accumulate more than 50 sugar, so this equation must absorb all ants whose sugar level would have reached 50 or higher:

$$w_{r,m,50}(t+1) = \sum_{i=50-(r-m)}^{50} w_{r,m,i}(t). \tag{5}$$

Similar modifications are necessary for equations tracking the number of ants with 0 sugar. The full system of equations is provided in "Appendix 2"; explanations here are meant to provide some insight as to how the equations are formulated.

Table 1 The order in which the equation system is updated

1. Update population equations based on movement
2. Store how many ants left each region
3. Scale wealth equations based on how many ants left that region
4. Update wealth equations based on eating and metabolism
5. Scale wealth equations based on taxation (if applicable)
6. Store the amount of tax collected
7. Store the number of deaths
8. Scale population equations based on sum of wealth equations per region (taxation causes some ants to die; population equations need to reflect this)

3.3 Migration and taxation

Wealth distributions are affected not only by eating and metabolism, but by migration and taxation as well. When ants move from one region to another, the wealth equations for each region are affected as the incoming (and outgoing) ants take their wealth with them. Furthermore, taxation causes additional wealth shifts, and these are also region-dependent. In order to account for this, the equations are updated in a certain order; the order is determined by the order of events in the ABM. Table 1 explains the order in which the population and wealth equations are evaluated. Note that wealth equations are affected by migration (step 3) and population equations are in turn affected by taxation via the wealth equations (step 8).

4 Results

4.1 Population and wealth data

Figure 3 compares data from the ABM and the equation model **without taxation**. Since the optimization problem concerns taxation by region, it is necessary that the dynamics match *at the regional level*; hence, the four regional populations are presented (from bottom to top, the data correspond to regions 1–4, respectively). Note that there does not appear to be any error propagation; that is, though the simulation time was fixed in this case, the equations ought to capture dynamics for arbitrary time horizons.

4.2 Similarity with respect to control

Recall that a primary feature of the equation model is the ability to use it to solve the optimization problem. Thus, while the equations appear to be a near-perfect match for population and wealth levels without taxation, it is necessary to examine how the equations hold up when subjected to taxation. To that end, 100 tax schedules were generated at random and evaluated in both the ABM and the equation model. Figure 4 provides an example taken from these 100 schedules (data is labeled according to the corresponding region); the results for the other tax schedules are similar. As in

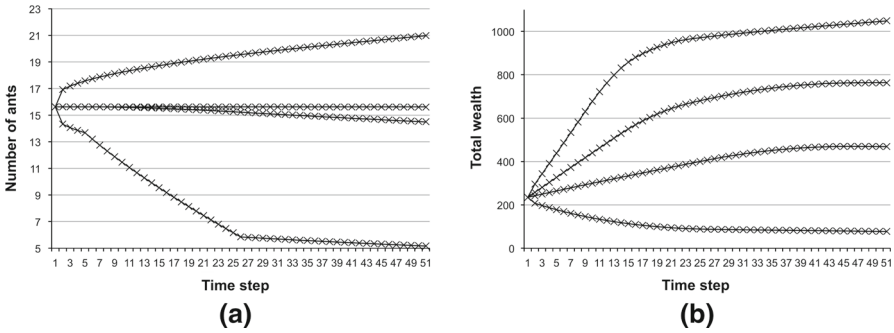

Fig. 3 Population and wealth data from the ABM (×) and the equation model (*lines*) (ABM data is averaged over 100 runs); from *bottom* to *top*, the data correspond to regions 1–4, respectively, **a** Population by region, **b** total wealth by region

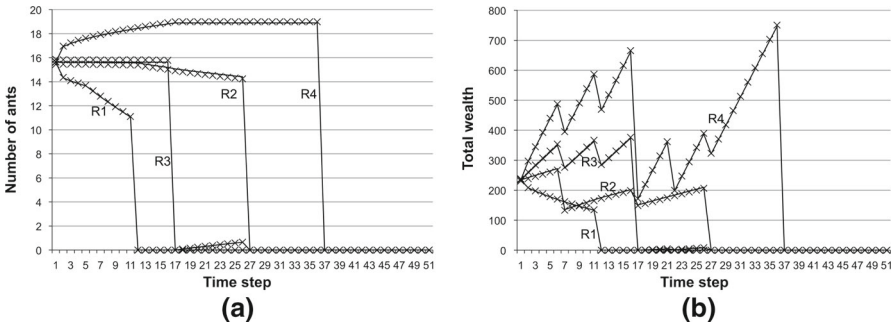

Fig. 4 Population and wealth data from the ABM (×) and the equation model (*lines*) for a randomly chosen tax structure (ABM data is averaged over 100 runs); data labels refer to the corresponding regions, **a** Population by region, **b** total wealth by region

the no-taxation case, the equations appear to capture the dynamics of the ABM nearly perfectly, further indicating that the equation model is a suitable surrogate—with respect to the optimization problem—for the ABM.

4.3 Pareto optimization

Having confirmed that the equation model truly captures the dynamics of the ABM with respect to the optimization problem, the equation model was used to solve the optimization problem. This was accomplished by a heuristic method known as *Pareto optimization*. Based on ideas first introduced by Italian economist Vilfredo Pareto in 1897 (Moore 1897), this evolutionary algorithm is designed for multiobjective optimization problems. The *Pareto frontier* consists of the set of solutions with the following property: no objective can be improved upon without some tradeoff in another. For our example, this means that the Pareto frontier consists of solutions wherein if we wish to obtain fewer deaths, we must use a tax schedule that results in lower income; alternately, if we wish to increase tax income, we can only do so with

schedules that cause more deaths. A typical evolutionary algorithm defines a fitness function that combines all of the objectives, perhaps by attaching weights to each, and then searches the solution space for the solution that maximizes fitness. Pareto optimization side-steps the issue of defining such a fitness function by searching for a set of solutions—each is optimal with respect to some choice of weights. There are several advantages to this approach: determining appropriate weights for a fitness function can be highly arbitrary, and may inadvertently focus attention on too narrow a region of the solution space. Additionally, heuristic algorithms typically examine many hundreds of solutions, but in the end return only one. By making use of a broader set of the solution space, Pareto optimization is more efficient—in fact, run time is similar to the case where only one solution is returned. A comprehensive review of heuristic methods for multiobjective optimization is outside the scope of this paper (for an extensive reference of other methods see Coello (2013)); Pareto optimization was chosen because it is intuitive, effective, and novel in its application to optimization for ABMs.

4.4 The algorithm

The Pareto optimization algorithm is based on a standard evolutionary algorithm: a population of solutions are generated at random and evaluated. Solutions from this population that contain advantageous features with respect to the optimization problem are then used to 'breed' subsequent solutions—child solutions are composed by selecting values from parent solutions at random. These solutions are then subjected to mutation, wherein taxation is either increased or decreased, in order to broaden the range of the Pareto frontier. The next generation of solutions consists of the Pareto optimal solutions from the prior generations plus the newly bred solutions. This process repeats for a fixed number of generations, or until some stopping condition is met. In this case, a solution is advantageous if it is either Pareto optimal with respect to the current generation of solutions, or if it has few neighbors in solution space. This latter trait rewards more unique solutions and encourages exploration of the solution space. Pseudo-code for the algorithm is provided in Algorithm 1.

4.5 Evolution of the Pareto frontier

There are two objectives in our optimization problem: minimizing death, and maximizing taxes collected. Each tax structure is evaluated in the equation model, and the values for these two objectives are stored; we can then visualize the solutions as data points on a scatter plot, where the axes correspond to the objectives. Examining the state of these plots at various times during the optimization algorithm gives some indication of its progress. Figure 5 provides four snapshots of the algorithm, taken from generations 0, 3, 8, and 25. Values from past generations remain on the plot in a lighter shade in order to accentuate the evolution of the solutions toward the Pareto frontier—as generations progress, we tend to see solutions with fewer deaths and higher income. A notable feature of the optimization algorithm is that by giving preference to solutions with fewer neighbors, a wide variety of Pareto optimal solu-

Algorithm 1 Pareto optimization pseudo-code.

```
generate random initial population of 40 solutions          ▷ each solution is a tax structure
while generation < 80 do
    evaluate current generation, call it current_pop          ▷ tally deaths and income
    determine Pareto frontier; call it current_frontier
    let next_pop = current_frontier                          ▷ save frontiers between generations
    while size of next_pop < size of generation_size do ▷ each generation contains a fixed number
        of solutions
        repeat
            choose two candidate solutions from current_pop
            let comp_set = random subset of 5 solutions from current_pop
            if exactly one candidate is Pareto dominant over comp_set then
                that candidate becomes parent          ▷ add candidates more likely to be on frontier
            else if one candidate has fewer neighbors in solution space then      ▷ give preference to
                isolated solutions
                that candidate becomes parent
            else
                choose candidate at random to become parent
            end if
        until two parents have been chosen
        breed two new solutions (a and b) using parents:
            for all components in parent solutions do                          ▷ uniform crossover
                select component from random parent
                add this component to a
                add corresponding component from other parent to b
            end for
            mutation_rate = 0.20((max_gens − generation)/max_gens)     ▷ mutation decreases
                over time
            for all components in a and b do                                   ▷ perform mutation
                change component with probability mutation_rate                ▷ permissible values:
                    {0, 0.25, 0.5, 0.75, 1}
            end for
        end routine
        add a and b to next_pop
    end while
    increase generation by 1
end while
```

tions are found. This is hinted at in Fig. 5d, where we see solutions spread across a range of values.

4.6 Validation

While the evolutionary features observed in Fig. 5 indicate that the algorithm is working, they do not serve as total validation of Pareto optimization. A better validation comes from the comparison of results from the algorithm with results from a random search, which serves as a baseline for heuristic methods. The algorithm was run for a total of 80 generations, each consisting of 40 solutions. Since the Pareto frontier carries over each generation, the total number of different schedules evaluated by the Pareto optimization algorithm was 821. Hence, 821 random solutions were generated and evaluated for comparison. The results are presented in Fig. 6. These results indicate that the Pareto optimization algorithm performs much better than a simple

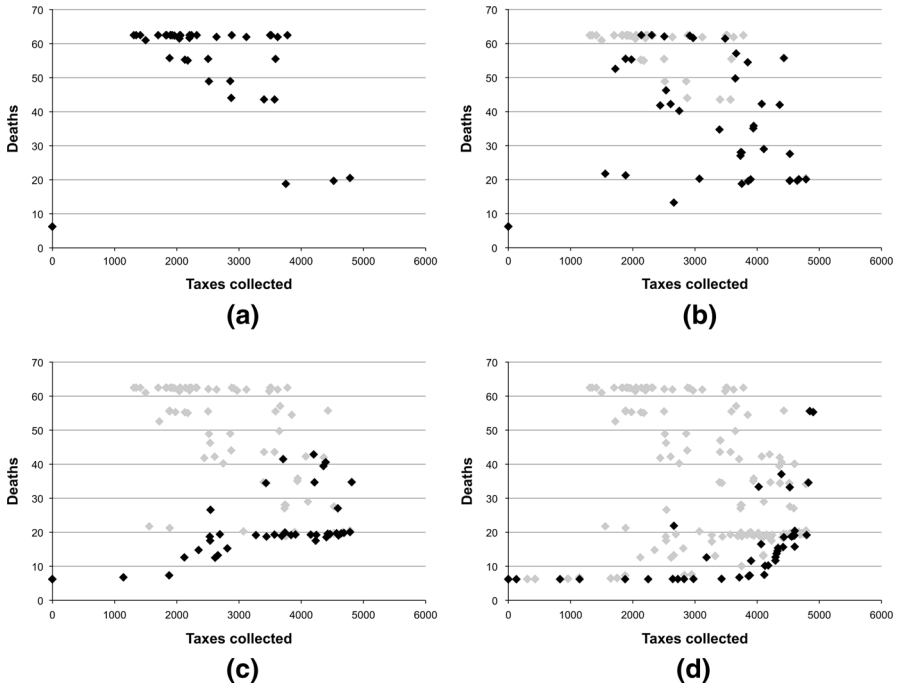

Fig. 5 Snapshots of the evolutionary Pareto optimization algorithm, **a** Gen. 0, **b** Gen. 3, **c** Gen. 8, **d** Gen. 25

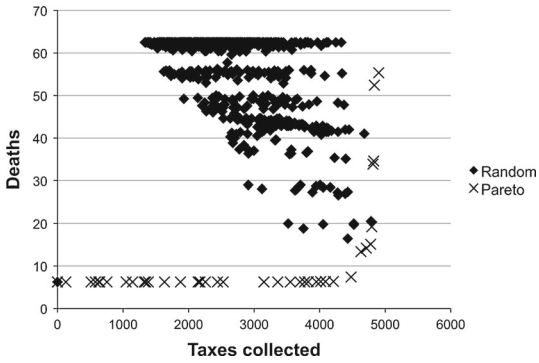

Fig. 6 A comparison of results from a random search and those from the Pareto optimization algorithm

random search. Note that a comprehensive overview of multi-objective optimization methods is beyond the scope of this text; however, this evaluation indicates the potential advantage of at least one such method. It is particularly worth noting that the algorithm returns a wide variety of solutions rather than just focusing on one region of the Pareto frontier. Another advantage of Pareto optimization is the insight offered into the choice of fitness function. The solution wherein no tax is imposed is a *de facto* member of the frontier since it is guaranteed to minimize deaths; however, as seen in Fig. 6, it is possible to obtain more than 4,000 sugar as tax without visibly

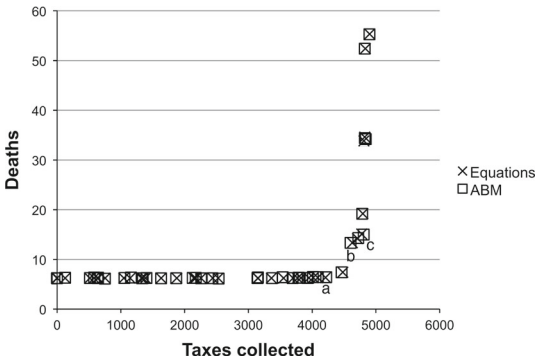

Fig. 7 The Pareto frontier evaluated in the equation model and in the ABM

increasing the number of deaths. On the other hand, the tradeoff between death and taxes increases drastically between 4,000 and 5,000 sugar collected, indicating that beyond 4,500 sugar or so, the tax can only be increased by a substantial sacrifice in the number of deaths. The shape of the frontier seems to suggest that for most reasonable fitness functions, there should be no more than 10 deaths and no less than 4,000 sugar collected.

The final step in the validation process is to check how the solutions obtained via the equation model actually track back to the ABM—after all, the equation model was built in order to aid investigation into an ABM optimization problem. To that end, the solutions comprising the final generation of the Pareto algorithm were implemented in the ABM; deaths and taxes were tallied for each. The results predicted by the equation model and those obtained via simulation in the ABM (with results averaged over multiple runs) are presented in Fig. 7. These results indicate that values obtained by the equation model are very similar to those obtained via ABM simulation, further validating both the qualitative and quantitative accuracy of the equations with respect to the dynamics of interest. It is worth noting that as taxation schedules approach 4,000 in terms of income, there is a sharp increase in the number of deaths. Taxation schedules for three such solutions (labeled *a*, *b*, and *c* in Fig. 7) are presented in Table 2. In these matrices, the rows correspond to regions 1–4 and the columns correspond to the ten tax cycles. Note that in all schedules, taxation is far less frequent in regions 1 and 2 than in regions 3 and 4. Additionally, taxation in regions 3 and 4 is more frequent in schedule *c*, and at generally higher rates. It is unclear whether the sharp increase in deaths that occurs at the phase transition is due more to an increase in frequency of taxation or to an increase in the actual rates; both are likely contributing factors. Examination of these tax structures provides some insight into real-world taxation considerations. While the model is indeed quite abstract, the goal of maximizing tax income while minimizing deaths has real-world implications. This may be easier to see if one thinks of 'death' as a sort of satisfaction metric. In that case, the results seem to indicate that periods of taxation ought to be punctuated by periods without taxation in order to better achieve both goals.

Table 2 Taxation schedules at the phase transition

Tax schedule	Taxation over time
a	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.25 & 0 & 0.25 & 0 & 0 & 0.25 \\ 0 & 0.5 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0.5 & 0.75 & 0 & 0 & 0.75 & 0.75 & 0 & 0.75 \\ 0 & 0.5 & 0 & 0.75 & 0.75 & 0.25 & 0.25 & 0 & 0.25 & 0.75 \end{bmatrix}$
b	$\begin{bmatrix} 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0.25 & 0.5 & 0 & 0.75 & 0.5 & 0.75 & 0.75 \\ 0 & 0.25 & 0.75 & 0.75 & 0 & 0 & 0 & 0.75 & 0 & 0.75 \\ 0.5 & 0.5 & 0 & 0.25 & 0.5 & 0.75 & 0 & 0 & 0.75 & 0.75 \end{bmatrix}$
c	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.25 & 0 & 0.5 & 0 & 0.75 & 0 \\ 0.5 & 0 & 0.25 & 0 & 0.75 & 0.5 & 0.5 & 0.5 & 0.75 & 0.75 \\ 0.25 & 0.5 & 0.75 & 0.75 & 0 & 0.75 & 0.75 & 0.75 & 0.75 & 0.75 \\ 0.5 & 0.5 & 0 & 0.75 & 0.75 & 0.75 & 0.25 & 0.5 & 0.75 & 0.75 \end{bmatrix}$

5 Discussion

In this paper, a spatially heterogeneous agent-based model is approximated by a system of discrete difference equations, which are then used to solve an optimization problem. The example presented here illustrates an approach to solving optimization problems for ABMs which ought to have wide appeal: ABMs are often created to investigate real-world systems, and optimization problems typically arise naturally in many such investigations. Here, the equation system was shown to capture pertinent ABM dynamics for a variety of conditions, and Pareto optimization was used as a novel means for solving the optimization problem. The results of this heuristic approach are encouraging, as they serve as an effective means for optimization when other theoretical approaches are not tractable in practice. Finally, the results obtained via the equations were shown to track faithfully back to the ABM, serving as a sort of a posteriori validation of the equation system. It should be noted that in any case, the results obtained are likely to be dependent on specific characteristics of the ABM, including number of agents, variability of agent traits, and the size of the system. While there may be cases for which this approach is not suitable, this example serves as evidence that equation-based analysis of ABMs holds the potential to be a powerful tool for a broad class of models.

5.1 Future work

In this case, the equation system was built analytically from the rules of the ABM. However, in many ABMs this approach is not practical: either the rules are too complicated to express mathematically, or the dynamics of interest rely on too many factors. In such cases, phenomenological equations must be developed that fit ABM data. While principles from symbolic regression can be applied, they are not well-established in

the framework of ABM analysis. This is an area of ongoing research and preliminary results are promising. It was mentioned earlier that there may be cases for which the approach presented here is inappropriate; hence, the ability to classify models based on susceptibility to equation-based analysis would be of particular interest. For any attempts at describing ABMs via equations, statistical methods need to be developed that can bring rigor to the discussion of model similarity; we are examining such methods in another ongoing project. Finally, a similar study to that presented here, using partial differential equations rather than difference equations, is currently underway. This will help contrast the advantages and disadvantages of the continuous and discrete modeling frameworks as applied to mathematical analysis of ABMs.

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6 Appendix 1: Overview, design concepts, and details (ODD) protocol for the SugarScape gradient model

This appendix contains the Overview, Design Concepts, and Details (ODD) protocol for the agent-based model examined in this paper. This protocol was introduced by Grimm et al. (2006, 2010) as a template for describing an agent-based model in detail sufficient enough that the model (and results) can be replicated by an independent researcher. The protocol also includes a list of modeling features that the ABM contains. The description in this appendix is meant to stand alone; as such, overlap may occur between the description here and that contained in the body of the article.

6.1 Purpose

The SugarScape Gradient model is a simplified version of SugarScape (Epstein and Axtell 1996), a model in which abstract entities (ants) roam a landscape made of sugar. Ants all have the same vision strength, but they have varying metabolisms, affecting the amount of sugar burnt over time. Ants that are able to make it to the sugar-rich regions on the right side of the map tend to survive, while those who do not are more likely to run out of sugar and die. Additionally, ants are periodically taxed for their sugar stores—tax rates may be 0, 25, 50, 75 %, or 100 %, and the rates are set regionally (hence regions with more sugar may have higher taxes, for example). Each tax cycle, tax rates may change. The purpose of this model is to investigate the effects of various taxation policies on tax income and ant population. In particular, the model is used to investigate the following question: what is the optimal taxation policy for maximizing collected income while minimizing deaths?

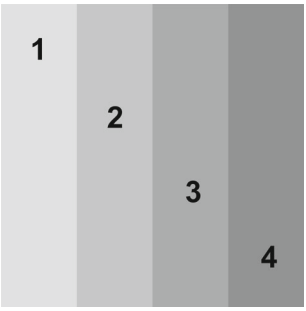

Fig. 8 Landscape of the SugarScape gradient model

Table 3 Taxation and temporal variables

Variable	Abbreviation	Value(s)	Units
Simulation duration	$total_sim_time$	51	Time steps
Tax rate	tax	$\{0,\ 0.25,\ 0.5,\ 0.75,\ 1\}$	N/A
Tax interval	$tax_interval$	5	Time steps

6.2 Entities, state variables, and scales

6.2.1 Grid cells, spatial, and temporal scales

The landscape is a 48×48 grid of cells containing various levels of sugar. Figure 8 presents a snapshot of the landscape; labels refer to the amount of sugar contained on each grid cell in that region. The landscape wraps vertically but **not** horizontally; thus, the only way to reach a region with more sugar is by traveling to the right. When ants consume the sugar from a grid cell, the sugar is not depleted. Hence sugar levels are preserved throughout each simulation. Given the fairly abstract nature of the model, time and space are unitless. A simulation consists of a finite number of time steps. At regular intervals, taxes are either collected or not, depending on the tax policy being simulated.

Taxation. Taxes are collected every 5 time steps for a total of 10 tax cycles, and tax rates may be different in each region and at each tax cycle. Hence, using the values in Table 3, a tax policy is a 4×10 matrix with entries in $\{0,\ 0.25,\ 0.5,\ 0.75,\ 1\}$. Entry (i, j) of this matrix indicates the tax rate in region i during the jth tax cycle. Thus there are a total of $5^{40} \approx 9 \times 10^{27}$ possible tax policies. For each policy, the resultant number of deaths and the amount of tax collected are stored.

6.2.2 Ants

Each ant has a fixed vision level of 1 grid cell for the duration of the simulation: this means ants can see exactly one cell in each of the four principal directions up, down, left, and right (unless they are on the horizontal edge of the map, in which case vision does not wrap). Additionally, each ant has a fixed metabolism which determines how

Table 4 Ant state variables

State variable	Abbreviation	Value(s)
Location (current region)	reg	$\{1, \ldots, 4\}$
Vision	vis	1
Metabolism	met	Random in $\{1, 2\}$
Sugar	sug	$\{1, 2, \ldots 50\}$

much sugar it loses each time step. Movement is governed by vision: an ant moves to the grid cell within its vision with the maximum amount of sugar. Multiple ants may occupy the same grid cell; each ant obtains the full amount of sugar at that location (i.e., sugar is not depleted and there is no exclusion principle in effect). Ants die if their sugar level reaches zero. Ants may accumulate up to 50 sugar (Table 4).

6.3 Process overview and scheduling

Per the suggestion in Grimm et al. (2010), the ABM process is presented in Algorithm 2 as pseudo-code in order to give a detailed account of the order of events. The ant and tax routines are executed fully by one ant, then fully by another—i.e., serially. This means that state variables are updated asynchronously. Time steps are discrete units, as is movement: ants jump directly from the center of one grid cell to the center of another.

Algorithm 2 Process pseudo-code.

```
setup map and constants                                   ▷ layout is read in from a matrix in a.txt file
while simulation time < total_sim_time do
    store population and wealth data
    ant routine:
        let vision_cells be set of grid cells within 1 grid cell of location      ▷ includes current cell
        let potential_cells be set of vision_cells with maximum sugar   ▷ maximum sugar cell may
            not be unique
        move to randomly selected member of potential_cells
        sug = sug − met + s_here                                          ▷ eat and metabolize sugar
    end routine
    if time mod tax_cycle = 0 then                        ▷ collect tax every tax_interval time steps
        for all ants do
            sug = sug − ⌈tax · sug⌉                                   ▷ taxes are always rounded up
        end for
        update amount paid by all ants                                  ▷ keep track of tax income
    end if
    ants with sugar ≤ 0 die
    store death data
end while
write all data to file
```

6.4 Design concepts

6.4.1 Basic principles

The SugarScape Gradient model builds on SugarScape, which can be studied in a myriad of contexts, including sociology, ecology, economics, and culture. The primary motivation for this version is to investigate the effects of various tax policies in an abstract setting. It is hoped that the model will provide insight into the relationship between tax policies and population dynamics, and that these insights might be extrapolated to some real-world system.

6.4.2 Emergence

Spatial population dynamics ought to be an emergent property of SugarTax: for example, frequent high taxation in high-sugar regions might substantially alter regional population counts. The precise mechanism driving such changes is not built in to the model in any direct sense.

6.4.3 Objectives

The objective of each ant is to move to a cell within its vision with the maximum amount of sugar. There is no other consideration, and ants do not have knowledge of past or future taxation in any location.

6.4.4 Sensing

Ants are aware of the sugar level and occupancy of each grid cell within their vision. They are not aware of any properties of any other ants, including those within their vision.

6.4.5 Stochasticity

Ant movement is partially stochastic: if there are multiple grid cells within an ants vision with the same (maximum) sugar level, the ant will move to one of these grid cells at random. This stochasticity is vital to population distributions—if the ants chose the nearest such cell, for example, populations tend to cluster on the boundary of each region. Clustering severely affects migrational patterns, as ants are not encouraged to explore the landscape. This movement feature is discussed in Epstein and Axtell (1996).

6.4.6 Observation

Each simulation occurs over a finite number of time steps. At each time step, the following information is collected: the number of ants in each region, the wealth distribution of the ants in each region (separated into disjoint bins), the total amount of tax collected, and the number of deaths occurring in each region. In addition, the

tax policy implemented over the course of the simulation is recorded as well. At the end of each simulation, these data are written to a comma separated value (.csv) file, a universal format for spreadsheet applications.

6.5 Initialization

The model is initialized with an average of 62.5 ants; each is placed at a random location on the landscape. Ants begin with a random amount of sugar between 5 and 25 (inclusive); this value is different for each ant and chosen from a uniform distribution. Ants are initialized with vision 1 and metabolism either 1 or 2. Vision and metabolism of a given ant do not change over the course of a simulation.

6.6 Input data

The landscape is read in from a.txt file; this helps with implementation and makes it easier to make changes. A tax policy can either be chosen directly via code manipulation or chosen at random. The policy must be chosen prior to simulation. As such, the tax policy may be thought of as input to the model.

7 Appendix 2: Difference equation models

Figure 9 contains the population equations for the 48 columns in the ABM; Fig. 10 contains the wealth (i.e., sugar level) equations. Note that in order to match ABM behavior, all equations are updated **synchronously**. See Table 1 for the order in which equations are updated. Note that since there are multiple updates of the equations in order to simulate one time step in the ABM, the time scale in the equation system is not quite the same. In particular, time steps in the equations are more closely aligned with *processes* in the ABM (for example, agent movement or taxation).

7.1 Initial values

Initial values for each of the equations are provided in Fig. 11. These values are determined by settings from the ABM. Note that in this study we focus only on

$$
\begin{aligned}
p_1(t+1) &= 0.75p_1(t) + 0.2p_2(t), \\
p_a(t+1) &= p_{a-1}(t) + 0.75p_a(t) + 0.2p_{a+1}(t), && a \in \{13, 25, 37\}, \\
p_b(t+1) &= 0.25p_{b-1}(t) + 0.6p_b(t) + 0.2p_{b+1}(t), && b \in \{2, 14, 26, 38\}, \\
p_c(t+1) &= 0.2p_{c-1}(t) + 0.6p_c(t) + 0.2p_{c+1}(t), && c \in \{3, \ldots, 10, 15, \ldots, 22, \\
& && 27, \ldots, 34, 39, \ldots, 46\}, \\
p_m(t+1) &= 0.2p_{m-1}(t) + 0.6p_m(t), && m \in \{11, 23, 35\}, \\
p_n(t+1) &= 0.2p_{n-1}(t), && n \in \{12, 24, 36\}, \\
p_{47}(t+1) &= 0.2p_{46}(t) + 0.6p_{47}(t) + 0.25p_{48}(t), \\
p_{48}(t+1) &= 0.2p_{47}(t) + 0.75p_{48}(t).
\end{aligned}
$$

Fig. 9 Population equations

$$
\begin{aligned}
w_{1,1,0}(t+1) &= 0, \\
w_{1,1,a}(t+1) &= w_{1,1,a}(t) && a \in \{1,\ldots,50\}, \\
w_{1,2,b}(t+1) &= w_{1,2,b+1}(t) && b \in \{0,\ldots,49\}, \\
w_{1,2,50}(t+1) &= 0, \\
w_{2,1,0}(t+1) &= 0, \\
w_{2,1,a}(t+1) &= w_{2,1,a-1}(t) && a \in \{1,\ldots 49\}, \\
w_{2,1,50}(t+1) &= w_{2,1,49}(t) + w_{2,1,50}(t), \\
w_{2,2,0}(t+1) &= 0, \\
w_{2,2,b}(t+1) &= w_{2,2,b}(t) && b \in \{1,\ldots,50\}, \\
w_{3,1,a}(t+1) &= 0, && a \in \{0,1\}, \\
w_{3,1,b}(t+1) &= w_{3,1,b-2}(t) && b \in \{2,\ldots,49\}, \\
w_{3,1,50}(t+1) &= w_{3,1,48}(t) + w_{3,1,49}(t) + w_{3,1,50}(t), \\
w_{3,2,0}(t+1) &= 0, \\
w_{3,2,c}(t+1) &= w_{3,2,c-1}(t) && c \in \{1,\ldots,49\}, \\
w_{3,2,50}(t+1) &= w_{3,2,49}(t) + w_{3,2,50}(t), \\
w_{4,1,a}(t+1) &= 0 && a \in \{0,1,2\}, \\
w_{4,1,b}(t+1) &= w_{4,1,b-3}(t) && b \in \{3,\ldots,49\}, \\
w_{4,1,50}(t+1) &= w_{4,1,47}(t) + w_{4,1,48}(t) + w_{4,1,49}(t) + w_{4,1,50}(t), \\
w_{4,2,m}(t+1) &= 0 && m \in \{0,1\}, \\
w_{4,2,n}(t+1) &= w_{4,2,n-2}(t) && n \in \{2,\ldots,49\}, \\
w_{4,2,50}(t+1) &= w_{4,2,48}(t) + w_{4,2,49}(t) + w_{4,2,50}(t).
\end{aligned}
$$

Fig. 10 Wealth equations. Note that these equations track sugar levels based on sugar eaten and sugar metabolized, but do not account for migration or taxation. Here, the number of ants with 0 sugar is always set to 0; this is done to simulate death. Note that according to Table 1, wealth values are updated after migration and population values are updated after taxation. This ensures that at the end of each time step, the sum of the population equations is always the same as the sum of the wealth equations. Tax rates are percentages, but a ceiling function is applied in order to ensure that an integer amount of tax is paid

$$
\begin{aligned}
p_n(0) &= 62.5/48 && n \in \{1,\ldots,48\}, \\
w_{r,m,a}(0) &= 0 && a \in \{0,\ldots,4\}, \\
w_{r,m,b}(0) &= 62.5/(4 \cdot 21) && b \in \{5,\ldots,25\}, \\
w_{r,m,c}(0) &= 0 && c \in \{26,\ldots,50\}.
\end{aligned}
$$

Fig. 11 Initial values. 48 population equations track the number of ants in each column. There are four regions and 21 possible starting sugar amounts (5–25 inclusive); the 62.5 initial agents are assumed to be distributed among these regional sugar levels uniformly. Wealth equations hold for $r \in \{1,\ldots,4\}$ and $m \in \{1,\ 2\}$

agents with low vision (i.e., vision 1) and low metabolism (metabolism 1 or 2). In the original ABM, there were three other categories of ants, total of 250 inital agents. Hence, since we are focusing on one category, the initial population values must sum to $250/4 = 62.5$. These are placed at random throughout the grid. Likewise, ants begin the simulation with an amount of sugar chosen uniformly randomly between

5 and 25 (inclusive); hence the number of ants in a region determines the expected number at each sugar level.

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